

ROI Analyses Electrodes (temporal lobe, excluding hippocampus) - Model 1 (Encode vs Recall)

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Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/Data/FaceNameTask/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Cortex")
ROIdata <- read.csv("BOLD_TL_model1.csv")

# Assign factors
ROIdata$$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$StimType <- factor(ROIdata$StimType)
ROIdata$TaskStage <- factor(ROIdata$TaskStage)
```

Sham Condition - TL

```
#
# One sample t-test
Sham_enc <- subset(ROIdata, StimType=="Sham" & (TaskStage=="Encode" | TaskStage=="Recall"))
type = "greater"
Sham_ost <- ddply(Sham_enc, .(TaskStage), t.test.plyr, "BOLD")
Sham_ost <- ost_FDR(Sham_ost)
Sham_ost
```

```
## TaskStage statistic p.value estimate
## 1 Encode 3.34287683536345 0.00170918397086305 0.13506135
## 2 Recall -0.433727765783098 0.665314563418434 -0.0209825
## conf.int1 conf.int2 nobs dof method alternative
## 1 0.0651996639866771 Inf 20 19 One Sample t-test greater
## 2 -0.104632964830723 Inf 20 19 One Sample t-test greater
## null.value sig FDR
## 1 0 ** 0.003418368
## 2 0 <NA> 0.665314563
```

All Stim Conditions - TL

Does stimulation impact BOLD activity during encode vs recall stages of the task in the left temporal

```
Sp_data<- subset(ROIdata, (StimType=="Sham" | StimType=="TI 1:1" | StimType=="TI 1:3"))
xtabs(~ TaskStage + StimType, Sp_data)
```

```
## StimType
## TaskStage Sham TI 1:1 TI 1:3
## Encode 20 20 20
## Recall 20 20 20
```

```
BA.median <- lmer(BOLD ~ StimType*TaskStage + (1|S_ID), data = Sp_data, REML = TRUE)
Anova(BA.median, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
## F Df Df.res Pr(>F)
## StimType 0.9704 2 95 0.3827
## TaskStage 19.8443 1 95 2.294e-05 ***
## StimType:TaskStage 1.4511 2 95 0.2394
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Plot
p.BA.TL <- ggplot(Sp_data, aes(x = StimType, y = BOLD, fill = StimType)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.2) +
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust=1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
```

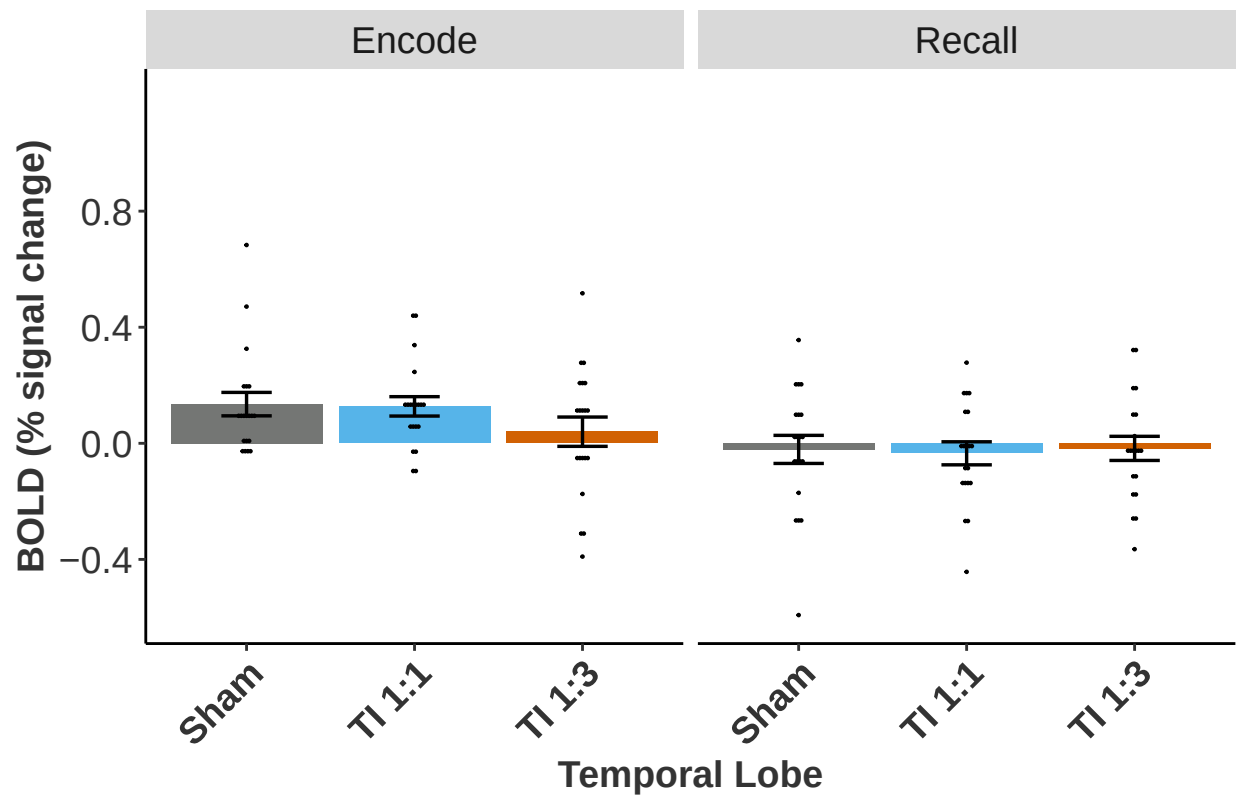
```

    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_x_discrete(breaks=c("Sham", "TI 1:1", "TI 1:3"),
    labels=c("Sham", "TI 1:1", "TI 1:3")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("") + xlab("Tempo

```

p.BA.TL

Bin width defaults to 1/30 of the range of the data. Pick better value with 'binwidth'.



```

ggsave(p.BA.TL, file="Figure_2Ri.tiff", width = 9, height = 9, dpi = 300, units = "cm", device = "tiff")

```

Bin width defaults to 1/30 of the range of the data. Pick better value with 'binwidth'.